Detailed Guide to Maximum Power Point Tracking (MPPT): Principles, Hardware Construction, and Implementation

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1 Introduction to MPPT

Maximum Power Point Tracking (MPPT) is a critical technique employed in renewable energy systems, particularly photovoltaic (PV) systems, to optimize the power extraction from variable power sources like solar panels or wind turbines. Solar panels produce power based on their voltage-current (I-V) characteristics, which are influenced by environmental factors such as solar irradiance, temperature, and shading. The goal of MPPT is to continuously adjust the operating point of the PV array to ensure it operates at the *maximum power point* (MPP), where the product of voltage and current ($P = V \times I$) is maximized.

Without MPPT, a PV system might operate at a suboptimal point, leading to significant energy losses (up to 30-50% in varying conditions). MPPT is typically implemented using power electronic converters (e.g., DC-DC converters) controlled by algorithms running on microcontrollers. This document provides a detailed explanation of MPPT principles, hardware construction for an MPPT charging system, the power vs. voltage (P-V) curve, common algorithms like Perturb and Observe (P&O) and Incremental Conductance (IncCond), and software implementations for STM32F103 or ATmega microcontrollers.

2 Principles of MPPT

The principle of MPPT revolves around the non-linear nature of PV cells. A PV cell can be modeled as a current source in parallel with a diode, with series and shunt resistances accounting for losses.

2.1 PV Cell Model and Characteristics

The equivalent circuit of a PV cell includes: - A photocurrent source (I_{ph}) proportional to irradiance. - A diode representing the p-n junction. - Series resistance (R_s) and shunt resistance (R_{sh}).

The output current (I) is given by:

$$I = I_{ph} - I_d - I_{sh}$$

where $I_d = I_0 \left(e^{\frac{V + IR_s}{nV_t}} - 1 \right)$ is the diode current, and $I_{sh} = \frac{V + IR_s}{R_{sh}}$.

The I-V curve is exponential, and the P-V curve is derived as $P = V \times I$, forming a hill-shaped curve with a single peak (MPP) under uniform conditions.

2.2 Factors Affecting MPP

- **Irradiance**: Higher sunlight increases current and power; MPP voltage shifts slightly. - **Temperature**: Higher temperature decreases voltage and power; MPP voltage drops. - **Shading/Load Variations**: Can create multiple local maxima, requiring global MPPT algorithms.

MPPT algorithms track the MPP by perturbing the operating point (e.g., via duty cycle adjustment in a converter) and observing power changes.

2.3 Common MPPT Algorithms

1. **Perturb and Observe (P&O)**: Perturbs voltage and checks power change. Simple but oscillates around MPP.

2. **Incremental Conductance (IncCond)**: Uses $dP/dV = 0$ at MPP, comparing dI/dV and $-I/V$. More accurate.

3. **Fractional Open-Circuit Voltage (FOCV)**: Approximates $V_{mpp} \approx k \times V_{oc}$ (k=0.7-0.8). Simple but inaccurate.

4. **Advanced**: Fuzzy logic, neural networks for complex scenarios.

3 Power vs. Voltage Curve

The P-V curve visually represents the power output as a function of voltage. For a typical 100W PV panel under standard test conditions (STC: 1000 W/mš, 25řC): - $V_{oc} = 22V$ - $I_{sc} = 6A$ - $V_{mpp} = 18V$ - $I_{mpp} = 5.56A$ - $P_{mpp} = 100W$

The curve peaks at the MPP and drops to zero at $V = 0$ and $V = V_{oc}$.

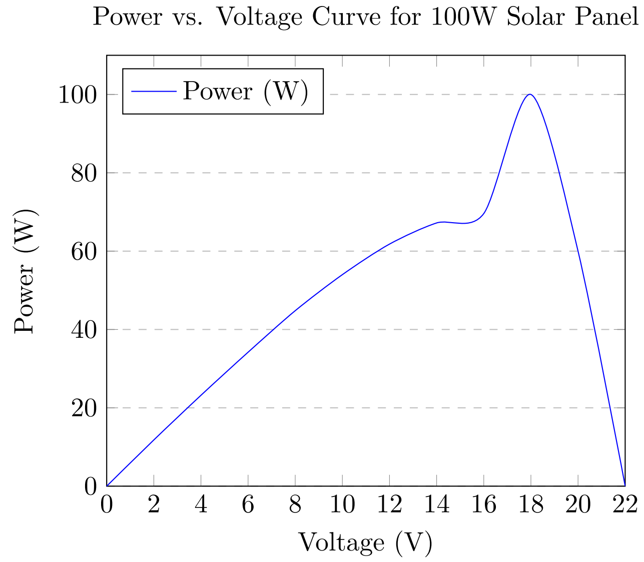

Figure 1: P-V Curve showing the MPP at 18V, 100W.

Table 1 lists the data points used for the plot.

Table 1: Power vs. Voltage Data Points

Voltage (V)	Power (W)
0	0
2	11.8
4	23.2
6	34.2
8	44.8
10	54.0
12	61.8
14	67.2
16	69.6
18	100.0
20	60.0
22	0

4 Constructing an MPPT Charging System: Hardware Side

Building an MPPT charging system involves integrating PV panels, a DC-DC converter, sensors, a microcontroller, and a battery/load. Below is a detailed guide.

4.1 Hardware Components

- **PV Panel**: Source of power (e.g., 100W, 18V MPP).
- **DC-DC Converter**: Adjusts voltage/current. Topologies:

- Buck: For $V_{out} < V_{in}$.
- Boost: For $V_{out} > V_{in}$ (common for battery charging).
- Buck-Boost: Flexible.

Components: MOSFETs (e.g., IRFZ44N), inductor (10-100 μH), capacitor (100-1000 μF), diode (Schottky for efficiency).

- **Microcontroller**: STM32F103 or ATmega328P for algorithm execution.
- **Sensors**:
  - Voltage: Divider circuit (resistors to scale to 0-3.3V/5V).
  - Current: Hall-effect (ACS712) or shunt resistor with op-amp (INA219).
- **Gate Driver**: For MOSFET switching (e.g., IR2110).
- **Battery/Load**: e.g., 12V/24V lead-acid or Li-ion with BMS.
- **Auxiliaries**: Filters, heatsinks, fuses for safety.

4.2 System Architecture

The PV panel connects to the DC-DC converter input. Sensors measure PV voltage/current. The microcontroller reads sensors via ADC, runs MPPT algorithm, and adjusts PWM duty cycle to control the converter. Output goes to battery/load.

4.2.1 Circuit Design Steps

1. **Select Topology**: Boost for stepping up to battery voltage.
2. **Calculate Components**: - Inductor: $L = \frac{V_{in}(V_{out}-V_{in})}{f \Delta I V_{out}}$, where f is switching frequency (20-50 kHz), ΔI ripple. - Capacitor: $C = \frac{I_{out} D}{f \Delta V}$, D duty cycle.
3. **Sensor Integration**: Voltage divider: $V_{out} = V_{pv} \frac{R_2}{R_1 + R_2}$.
4. **PWM Generation**: Microcontroller timer for high-frequency PWM.
5. **Protection**: Overvoltage (zener diodes), overcurrent (fuses), reverse polarity.

4.2.2 Example Boost Converter Schematic

(Conceptual: PV+ to inductor, inductor to MOSFET drain and diode anode, diode cathode to output capacitor and load. MOSFET gate from gate driver, controlled by PWM.)

4.3 Assembly and Testing

1. Prototype on breadboard/perfboard.
2. Simulate in LTspice/PSIM.
3. Test with variable resistor/load and light source.
4. Measure efficiency: $\eta = \frac{P_{out}}{P_{in}} > 95\%$.
5. Add enclosure for outdoor use.

5 Software Implementation: Incremental Conductance Algorithm

See Section 3 for theory. Below is the pseudocode for IncCond.

```
// Pseudocode for IncCond on Microcontroller
float Vpv, Ipv, Vpv_prev = 0, Ipv_prev = 0;
float duty_cycle = 0.5;
float step_size = 0.01;
float tolerance = 0.01;

void MPPT_IncCond() {
    Vpv = read_adc_voltage();
    Ipv = read_adc_current();
    float dV = Vpv - Vpv_prev;
    float dI = Ipv - Ipv_prev;

    if (dV != 0) {
        float inc_cond = dI / dV;
        float inst_cond = -Ipv / Vpv;
        if (fabs(inc_cond - inst_cond) < tolerance) {
            // At MPP
        } else if (inc_cond > inst_cond) {
            duty_cycle -= step_size;  // Increase voltage
        } else {
            duty_cycle += step_size;  // Decrease voltage
        }
    } else {
        if (dI > 0) duty_cycle -= step_size;
        else if (dI < 0) duty_cycle += step_size;
    }

    duty_cycle = constrain(duty_cycle, 0.0, 1.0);
    set_pwm(duty_cycle);

    Vpv_prev = Vpv;
    Ipv_prev = Ipv;
}

int main() {
    init_adc_pwm();
    while(1) {
        MPPT_IncCond();
        delay(100);  // ms
    }
}
```

5.1 Microcontroller-Specific Details

STM32F103: - ADC: 12-bit, dual channels. - PWM: TIM1, 72 MHz clock. - IDE: STM32CubeIDE.

ATmega328P: - ADC: 10-bit. - PWM: Timer1, 16 MHz clock. - IDE: Arduino.

6 Challenges and Advanced Topics

- **Partial Shading**: Use global MPPT. - **Efficiency Optimization**: Synchronous rectification. - **Integration**: With inverters for grid-tie systems. - **Resources**: IEEE papers, datasheets (e.g., STMicro for STM32).

7 Conclusion

This guide covers MPPT principles, hardware construction, and implementation. For practical builds, consult safety standards and simulate designs.